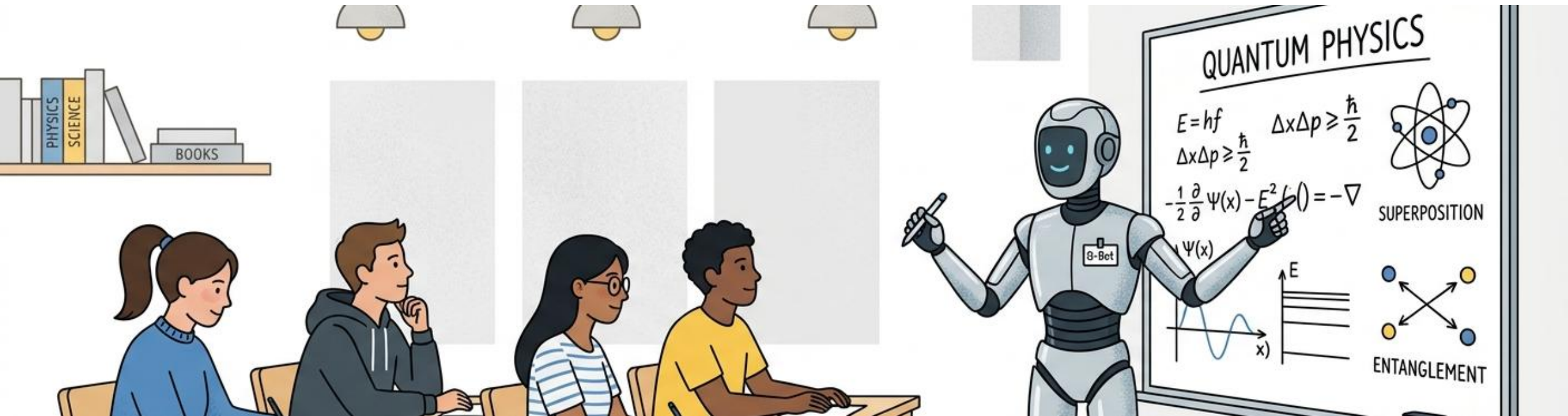
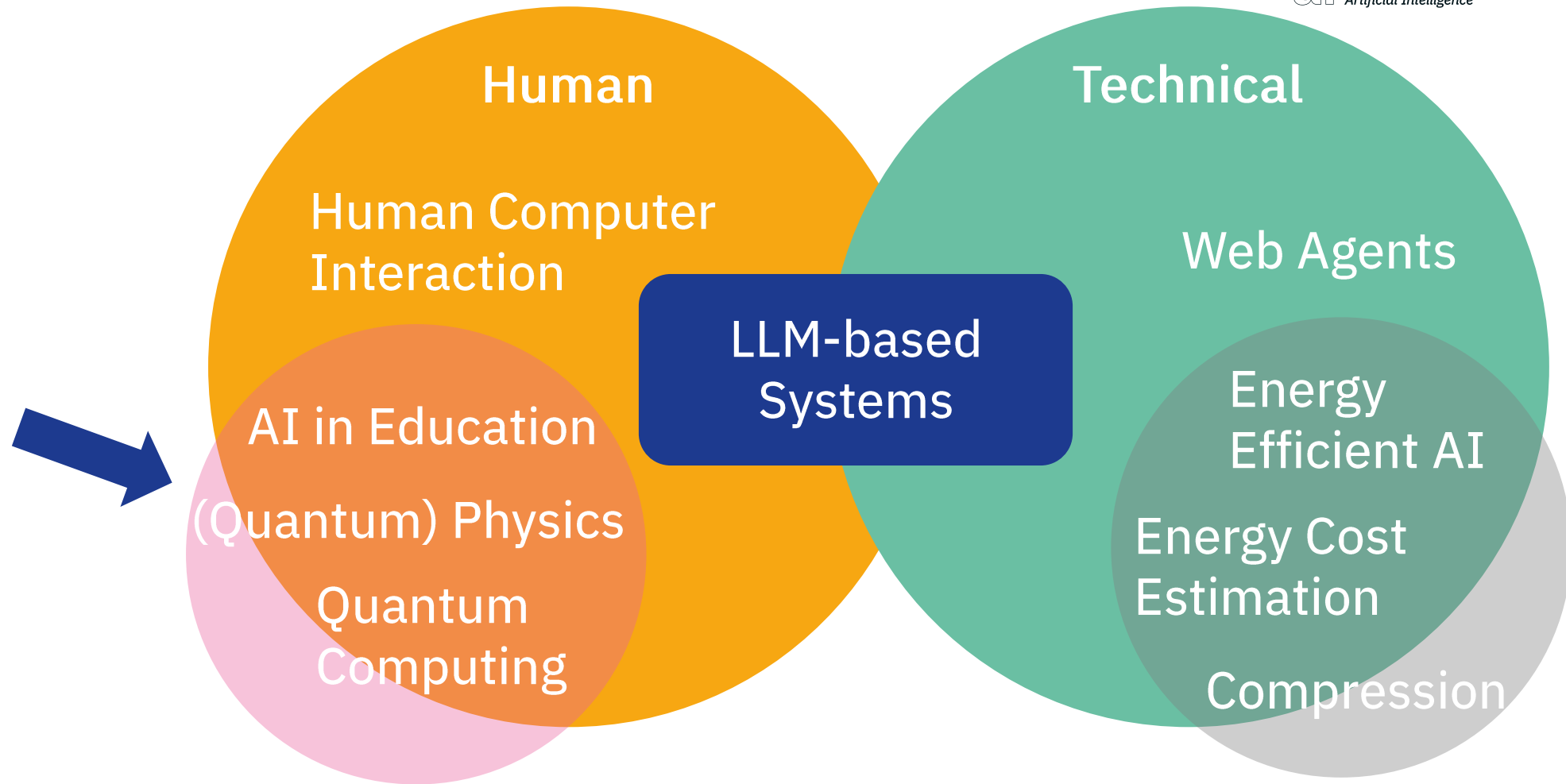




Challenges of GenAI in (Quantum) Education

Lars Krupp



Lars Krupp
PhD Student & Researcher @ DFKI & RPTU



The Advent of LLMs is Fundamentally Changing Education



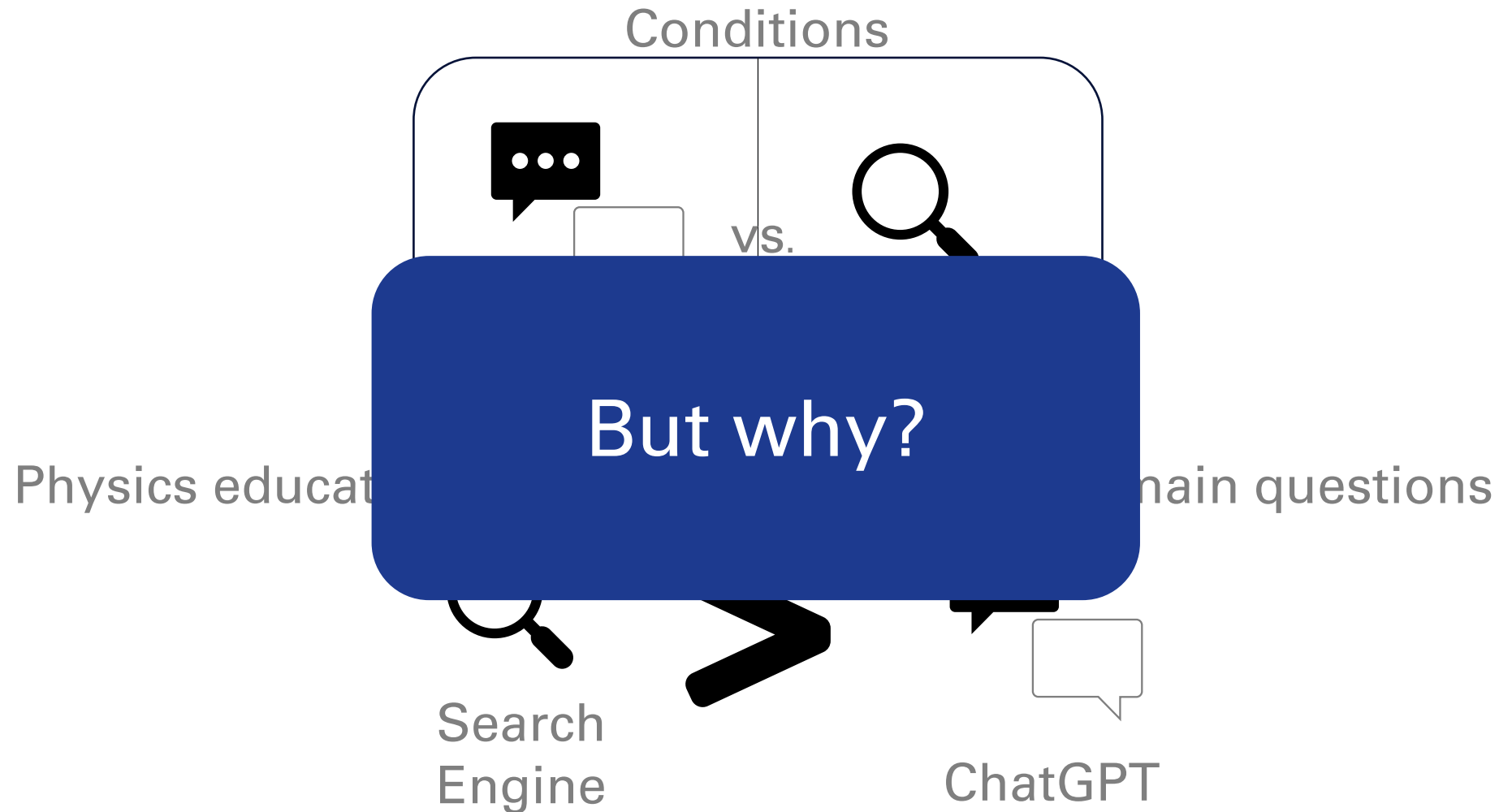
Initial Investigation

How does the use of ChatGPT influence performance?

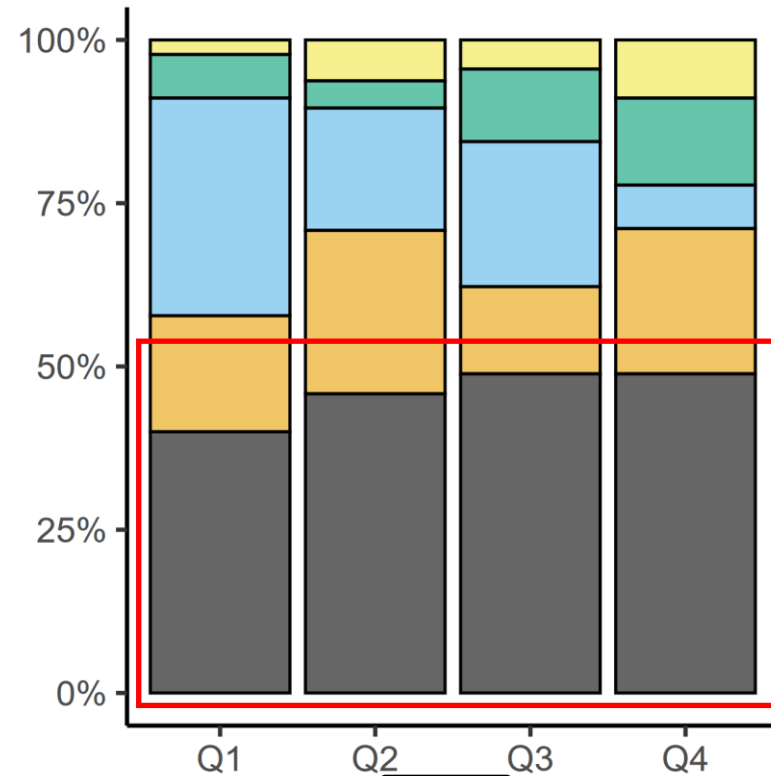
What are predominant interaction strategies?

Krupp, Lars, et al. "Unreflected acceptance—investigating the negative consequences of ChatGPT-assisted problem solving in physics education." *HHAi 2024: Hybrid Human AI Systems for the Social Good*. IOS Press, 2024. 199-212.

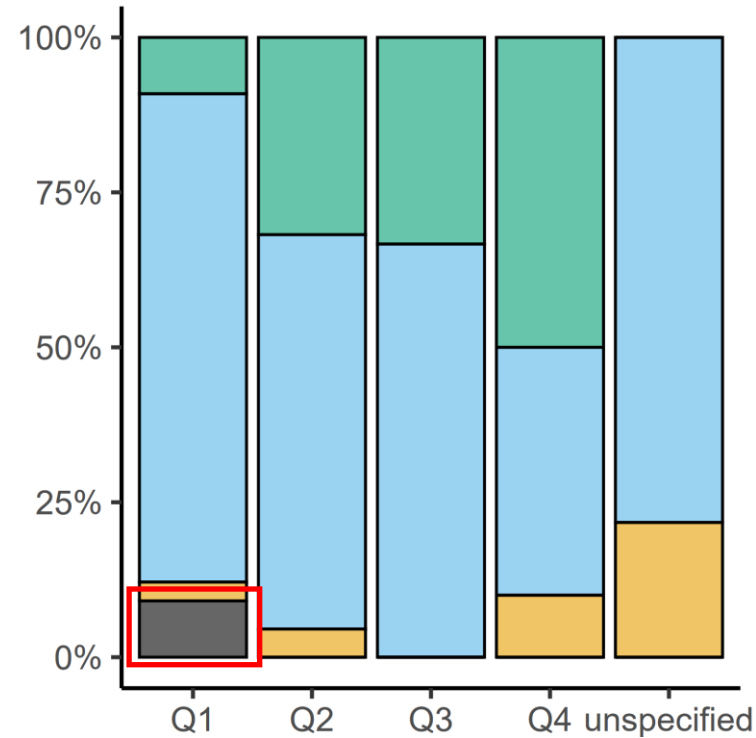
Student Performance



Interaction Strategies



ChatGPT



Search
Engine



Interaction Types

- Copy & Paste
- Postprocessing
- Preprocessing
- Transformation
- No Answer

Takeaways

Overreliance on
ChatGPT
answers leads to
low scores

Copy & Paste is
the most
prominent
strategy

What can we do?

Moderate LLM outputs
to mitigate the loss of
reflection



Hint Bot

Moderation
Conditions



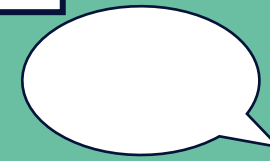
Multi-response
Bot



Search
Engine



Classic Bot



Baseline
Conditions

Krupp, Lars, et al. "Challenges and opportunities of moderating usage of large language models in education." *Proceedings of the International Conference on Mobile and Ubiquitous Multimedia*. 2024.

Hint Bot

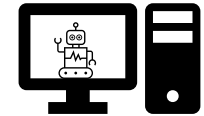
How do I calculate the length of the hypotenuse of a triangle without a right angle?



Calculate c if
 $a = 5$, $b = 7$
and $C = 60^\circ$

To calculate the length of the hypotenuse of a triangle without a right angle, you can use the Law of Cosines. The formula is $c^2 = a^2 + b^2 - 2ab \cdot \cos(C)$ [...]

To calculate the length of the hypotenuse (c) using the law of cosines, you can use the formula: [...]



Multi-response Bot

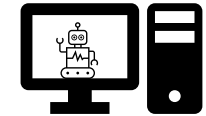
How do I calculate the length of the hypotenuse of a triangle without a right angle?



To calculate the length of the hypotenuse without a right angle, you can use the Pythagorean theorem or trigonometric functions.

To calculate the length of the hypotenuse of a triangle without a right angle, you can use the Law of Cosines.

You cannot calculate the length of the hypotenuse without a right angle.

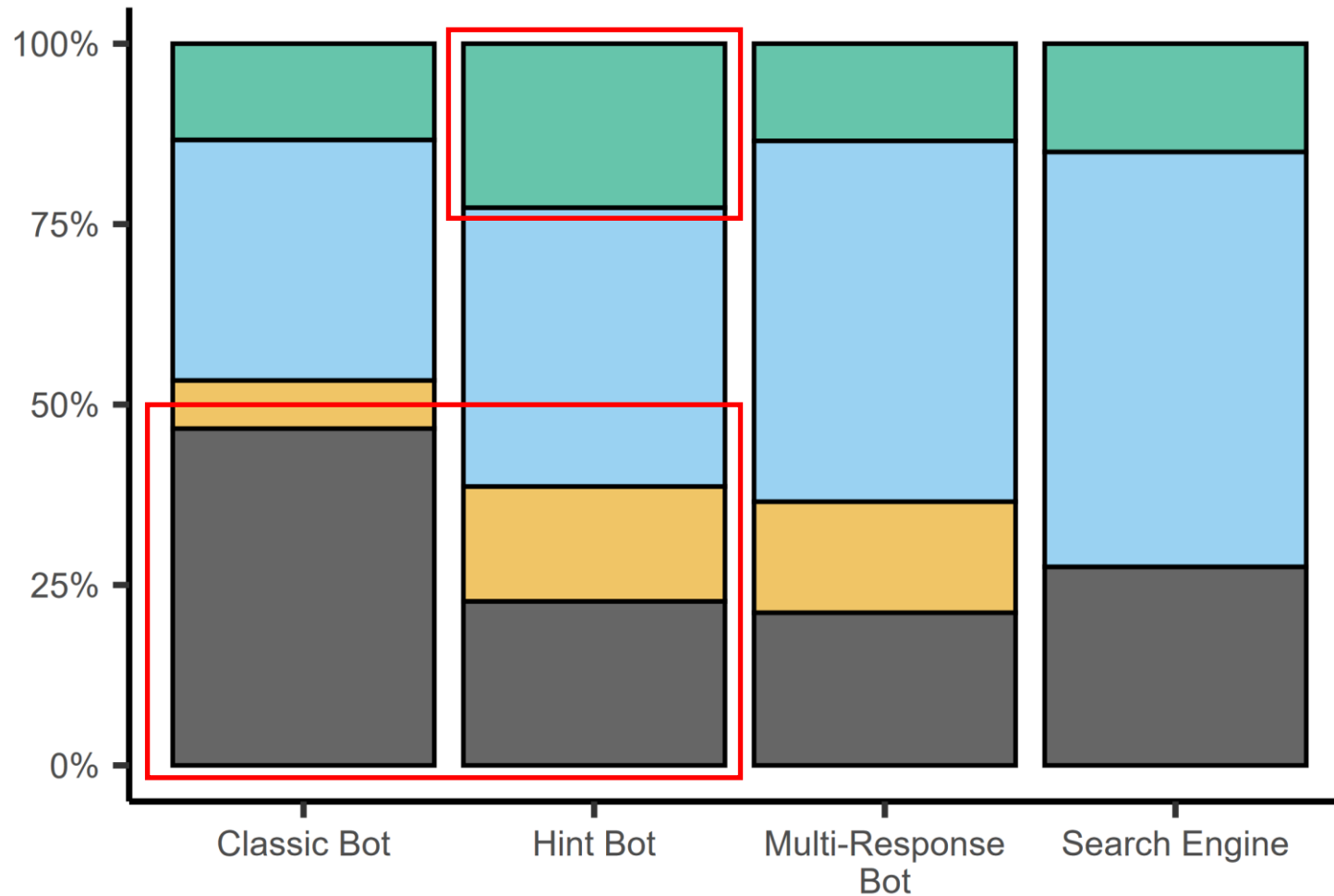


Student Performance

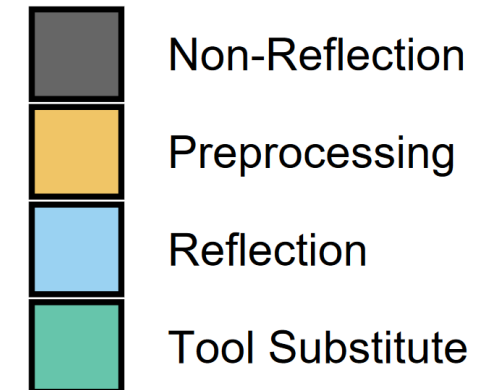
Physics educators scored the answers to the main questions



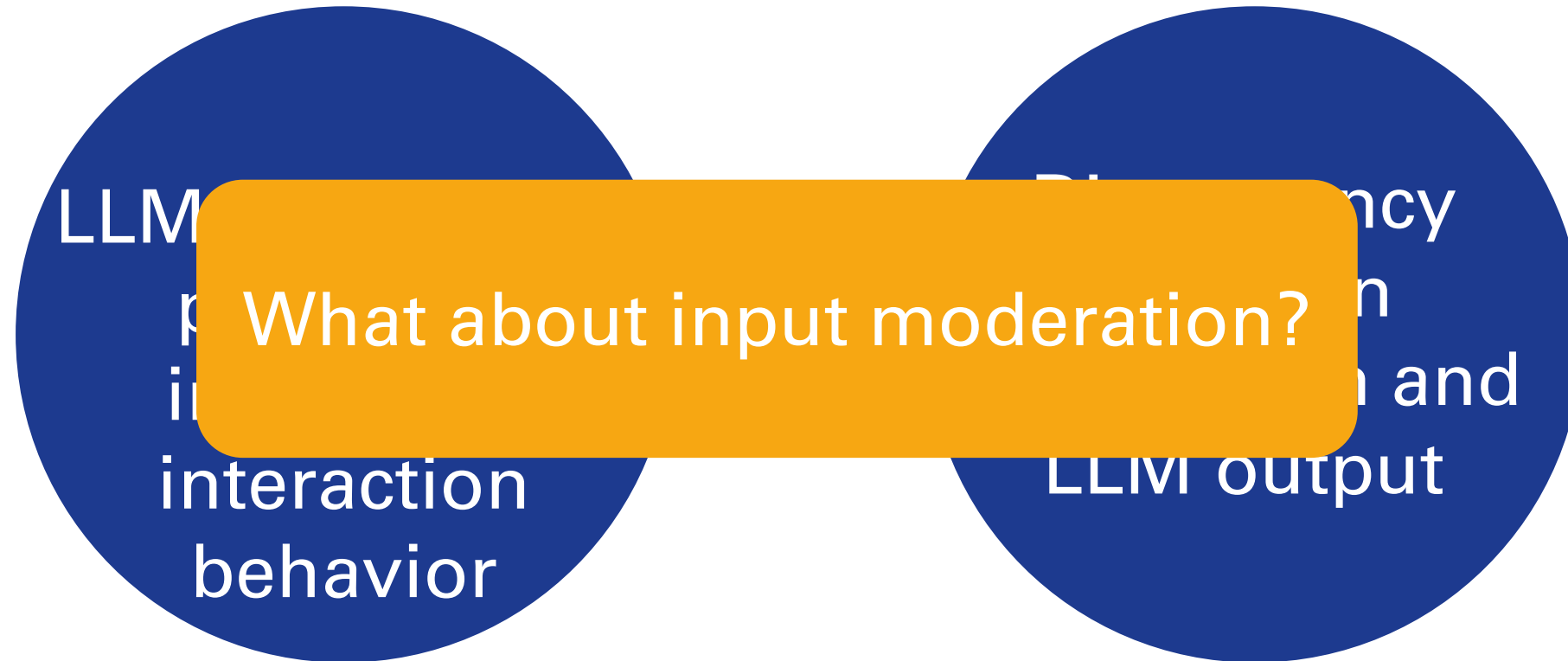
Interaction Types



Interaction Strategies

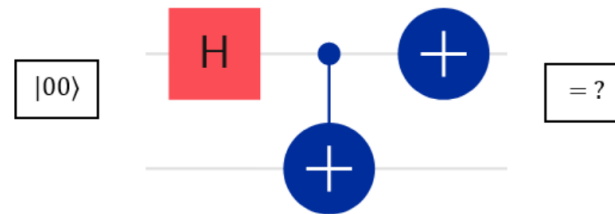


What we learned

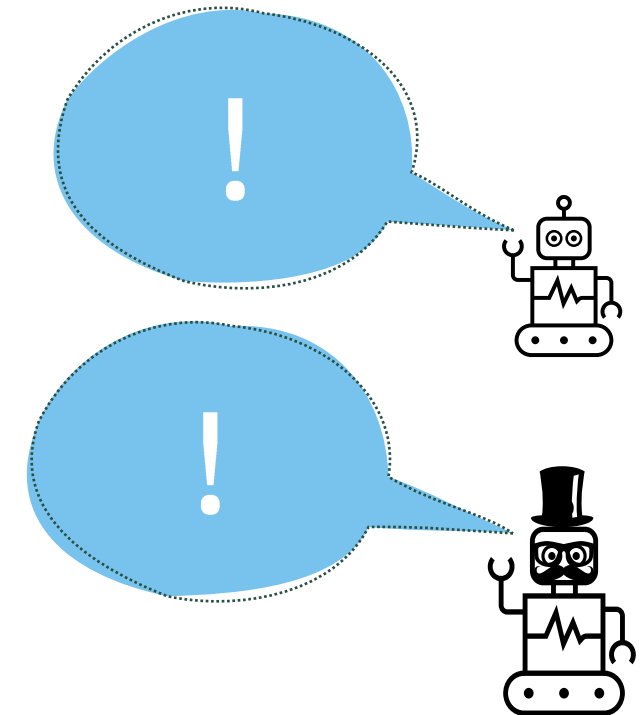


How do LLM-generated and expert-created Tips compare?

Which Bell-state is the outcome of the following operation on the $|00\rangle$ state: $(X \otimes I) \text{ CNOT}((H \otimes I)|00\rangle)$?



- ☐ a) $\frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$
- ☐ b) $\frac{1}{\sqrt{2}} (|00\rangle - |11\rangle)$
- ☐ c) $\frac{1}{\sqrt{2}} (|01\rangle + |10\rangle)$
- ☐ d) $\frac{1}{\sqrt{2}} (|01\rangle - |10\rangle)$

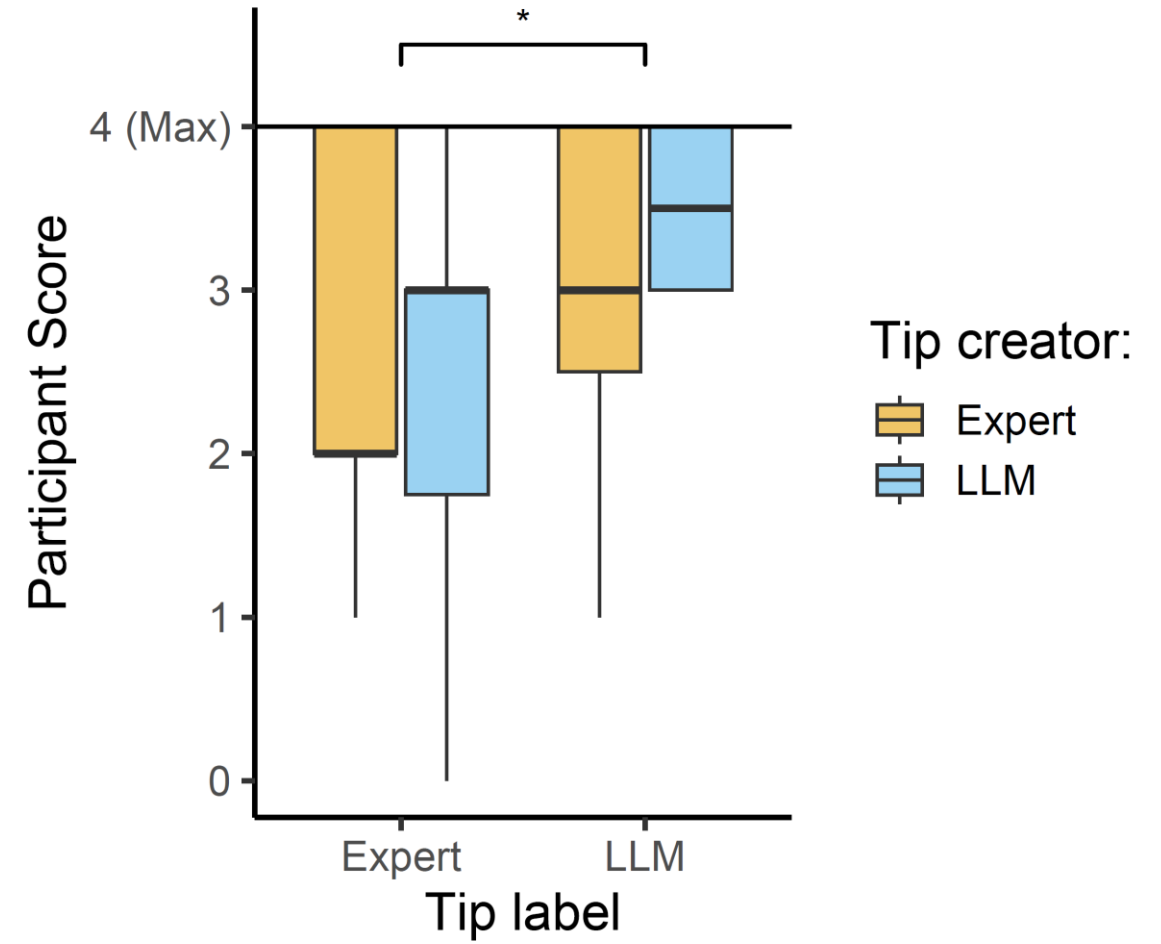


Krupp, Lars, et al. "LLM-generated tips rival expert-created tips in helping students answer quantum-computing questions." *EPJ Quantum Technology* 12.1 (2025): 33.

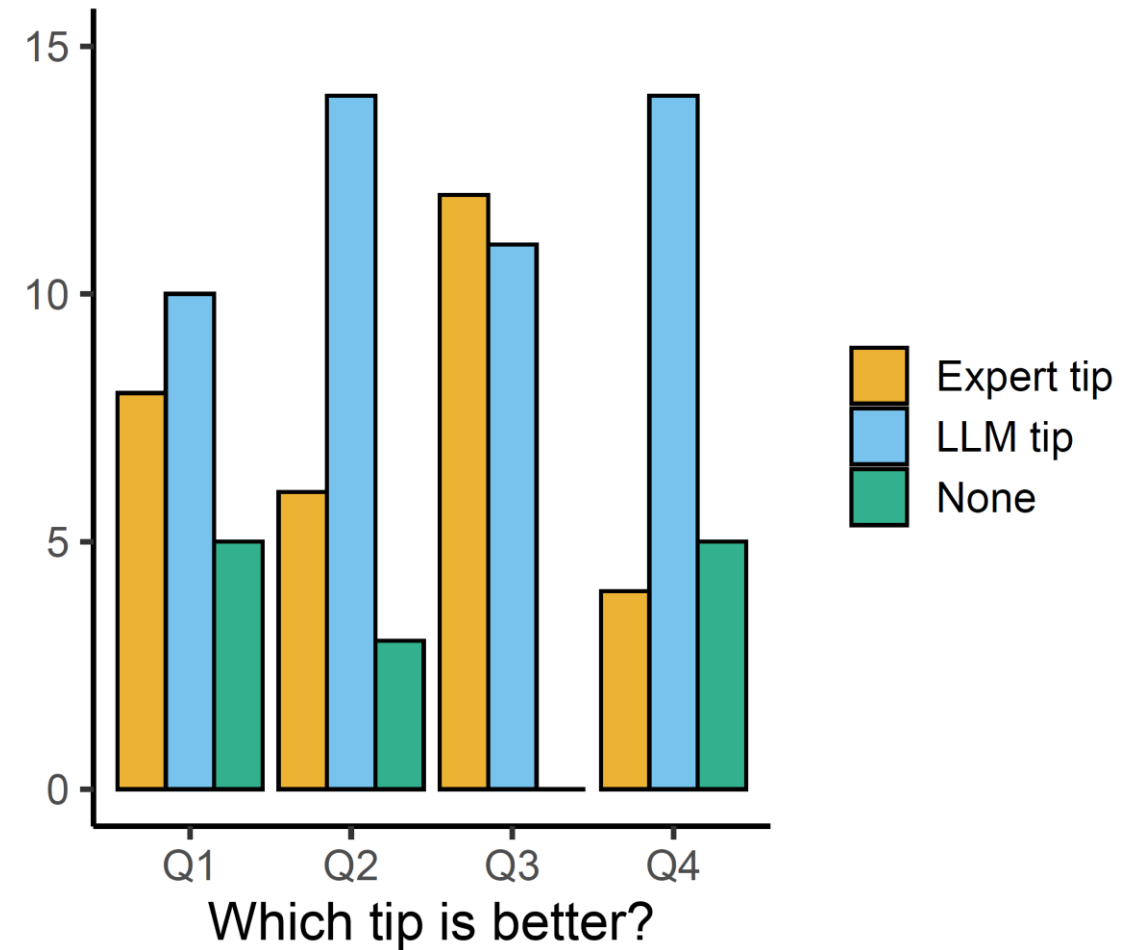
Results

Conditions

LLM Label	LLM Label
LLM Tip	Expert Tip
Expert Label	Expert Label
LLM Tip	Expert Tip



Results



What we learned



Take what we
learned



Chrome Plugin

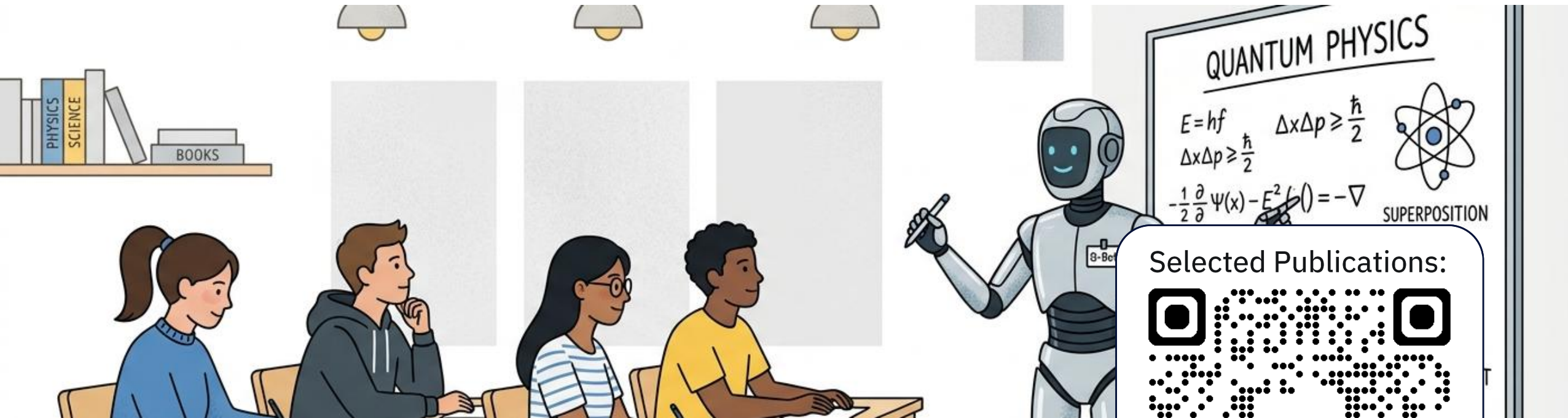
PENNYLANE

New Study on
Quantum
Computing
Education



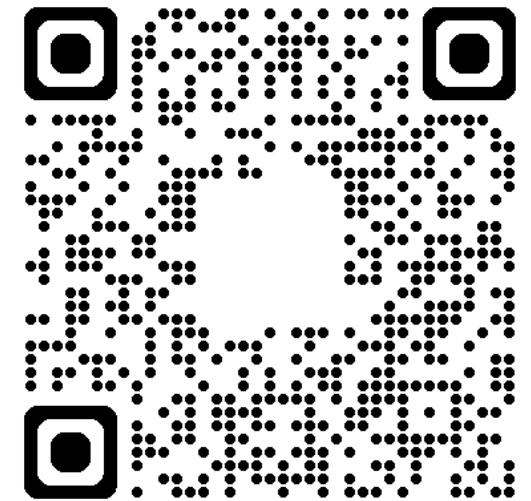
https://migroup.qualtrics.com/jfe/form/SV_5joIi0EykwMsbqe





Any Questions?

Selected Publications:



lars.krupp@dfki.de

